

CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

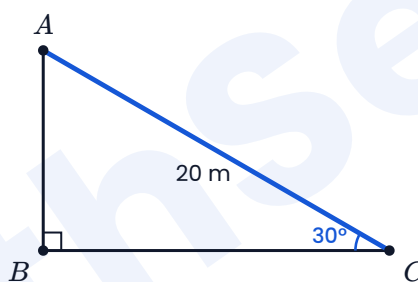
Chapter 9: Some Applications of Trigonometry

Step-by-step solutions to the textbook exercises: 15 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 9.1

- Q1** A 20 m rope is stretched tight from the top of a vertical pole to the ground and makes an angle of 30° with the ground. Find the height of the pole.



SOLUTION

Let AB be the pole and AC the rope. In right triangle ABC , $\sin 30^\circ = \frac{AB}{AC}$.

$$\frac{1}{2} = \frac{AB}{20}, \text{ so } AB = 10 \text{ m.}$$

Answer: 10 m

- Q2** A tree breaks in a storm and its top touches the ground 8 m from the foot of the tree, making an angle of 30° with the ground. Find the height of the tree.

SOLUTION

Let the tree break at a height h , and let the broken part have length l . The standing part, the ground and the broken part form a right triangle with base 8 m.

$$\tan 30^\circ = \frac{h}{8} \text{ gives } h = \frac{8}{\sqrt{3}} \text{ m. } \cos 30^\circ = \frac{8}{l} \text{ gives } l = \frac{16}{\sqrt{3}} \text{ m.}$$

$$\text{Height of the tree} = h + l = \frac{24}{\sqrt{3}} = 8\sqrt{3} \text{ m.}$$

Answer: $8\sqrt{3}$ m (about 13.86 m)

- Q3** Two slides are planned in a park: one whose top is 1.5 m high and inclined at 30° to the ground, and a steeper one whose top is 3 m high and inclined at 60° . Find the length of each slide.

SOLUTION

In each case the slide is the hypotenuse and the height is the side opposite the angle, so

$$\text{length} = \frac{\text{height}}{\sin \theta}.$$

$$\text{First slide: } \frac{1.5}{\sin 30^\circ} = \frac{1.5}{1/2} = 3 \text{ m. Second slide: } \frac{3}{\sin 60^\circ} = \frac{3 \times 2}{\sqrt{3}} = 2\sqrt{3} \text{ m.}$$

Answer: 3 m and $2\sqrt{3}$ m

- Q4** From a spot on level ground 30 m away from the base of a tower, the top of the tower is seen at an elevation of 30° . How tall is the tower?

SOLUTION

$$\tan 30^\circ = \frac{h}{30}, \text{ so } h = \frac{30}{\sqrt{3}} = 10\sqrt{3} \text{ m.}$$

Answer: $10\sqrt{3}$ m

- Q5** A kite flies 60 m above the ground. Its string is tied to a point on the ground and makes an angle of 60° with the ground. Find the length of the string, assuming it is straight.

SOLUTION

$$\sin 60^\circ = \frac{60}{l}, \text{ so } l = \frac{60 \times 2}{\sqrt{3}} = 40\sqrt{3} \text{ m.}$$

Answer: $40\sqrt{3}$ m

- Q6** A boy 1.5 m tall stands some distance from a 30 m tall building. As he walks towards it, the angle of elevation of its top from his eyes increases from 30° to 60° . How far did he walk?

SOLUTION

The top of the building is $30 - 1.5 = 28.5$ m above his eye level.

$$\text{At } 30^\circ \text{ his distance from the building is } \frac{28.5}{\tan 30^\circ} = 28.5\sqrt{3} \text{ m. At } 60^\circ \text{ it is } \frac{28.5}{\tan 60^\circ} = \frac{28.5}{\sqrt{3}} \text{ m.}$$

$$\text{Distance walked} = 28.5\sqrt{3} - \frac{28.5}{\sqrt{3}} = \frac{28.5 \times 2}{\sqrt{3}} = \frac{57}{\sqrt{3}} = 19\sqrt{3} \text{ m.}$$

Answer: $19\sqrt{3}$ m

- Q7** A transmission tower stands on top of a 20 m tall building. From one point on the ground, the bottom of the tower is seen at an elevation of 45° and its top at 60° . Work out the height of the tower.

SOLUTION

Let the point be d m from the building and the tower be h m high.

$$\tan 45^\circ = \frac{20}{d} \text{ gives } d = 20. \tan 60^\circ = \frac{20 + h}{d} \text{ gives } 20 + h = 20\sqrt{3}.$$

Answer: $h = 20(\sqrt{3} - 1)$ m (about 14.64 m)

- Q8** A 1.6 m tall statue is mounted on a pedestal. An observer on the ground sees the top of the statue at an elevation of 60° and the top of the pedestal at 45° . How high is the pedestal?

SOLUTION

Let the pedestal be h m high and the point be d m away. $\tan 45^\circ = \frac{h}{d}$ gives $d = h$.

$$\tan 60^\circ = \frac{h + 1.6}{h} \text{ gives } h\sqrt{3} = h + 1.6, \text{ so } h = \frac{1.6}{\sqrt{3} - 1} = \frac{1.6(\sqrt{3} + 1)}{2} = 0.8(\sqrt{3} + 1).$$

Answer: $0.8(\sqrt{3} + 1)$ m (about 2.19 m)

- Q9** Standing at the foot of a tower, you see the top of a building at an elevation of 30° . Standing at the foot of the building, you see the top of the tower at 60° . If the tower is 50 m tall, how tall is the building?

SOLUTION

Let the distance between them be d m and the building be h m high.

$$\tan 60^\circ = \frac{50}{d} \text{ gives } d = \frac{50}{\sqrt{3}}. \tan 30^\circ = \frac{h}{d} \text{ gives } h = \frac{d}{\sqrt{3}} = \frac{50}{3}.$$

Answer: $16\frac{2}{3}$ m

- Q10** Two poles of equal height stand on opposite sides of a road 80 m wide. From a point on the road between them, the angles of elevation of their tops are 60° and 30° . Find the height of the poles and the distances of the point from them.

SOLUTION

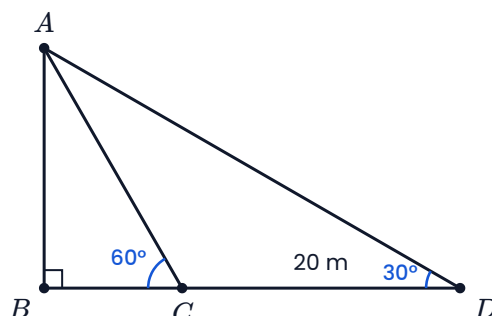
Let each pole be h m high and the point be x m from the pole seen at 60° , so $(80 - x)$ m from the other.

$$\tan 60^\circ = \frac{h}{x} \text{ gives } h = \sqrt{3}x. \tan 30^\circ = \frac{h}{80 - x} \text{ gives } h = \frac{80 - x}{\sqrt{3}}.$$

So $3x = 80 - x$, $x = 20$ and $h = 20\sqrt{3}$.

Answer: the poles are $20\sqrt{3}$ m high, and the point is 20 m and 60 m from them

- Q11** A TV tower stands on one bank of a canal. From the point directly opposite on the other bank the angle of elevation of its top is 60° . From a point 20 m further back on the same line the angle is 30° . Find the height of the tower and the width of the canal.



SOLUTION

Let $AB = h$ be the tower and $BC = x$ the width of the canal, with $CD = 20$ m.

In $\triangle ABC$: $\tan 60^\circ = \frac{h}{x}$, so $h = \sqrt{3}x$. In $\triangle ABD$: $\tan 30^\circ = \frac{h}{x+20}$, so $x+20 = \sqrt{3}h = 3x$.

So $x = 10$ and $h = 10\sqrt{3}$.

Answer: the tower is $10\sqrt{3}$ m high and the canal is 10 m wide

- Q12** A building is 7 m tall. From its roof, the top of a cable tower is seen at an elevation of 60° and the base of the tower at a depression of 45° . Find the height of the cable tower.

SOLUTION

Let the horizontal distance be d m. The angle of depression of the foot is 45° , so $\tan 45^\circ = \frac{7}{d}$ and $d = 7$.

The part of the tower above the level of the building top is $d \tan 60^\circ = 7\sqrt{3}$ m.

Answer: $7 + 7\sqrt{3} = 7(\sqrt{3} + 1)$ m (about 19.12 m)

- Q13** From the top of a lighthouse 75 m above sea level, the angles of depression of two ships are 30° and 45° . One ship is exactly behind the other on the same side. Find the distance between the ships.

SOLUTION

The ship seen at 45° is $\frac{75}{\tan 45^\circ} = 75$ m from the foot of the lighthouse. The ship seen at 30° is $\frac{75}{\tan 30^\circ} = 75\sqrt{3}$ m away.

Answer: $75\sqrt{3} - 75 = 75(\sqrt{3} - 1)$ m (about 54.9 m)

Q14 A girl 1.2 m tall sees a balloon moving in a horizontal line at a height of 88.2 m. The angle of elevation of the balloon from her eyes is 60° and, some time later, 30° . Find the distance travelled by the balloon in that time.

SOLUTION

The balloon is $88.2 - 1.2 = 87$ m above her eye level.

Horizontal distance at 60° : $\frac{87}{\tan 60^\circ} = \frac{87}{\sqrt{3}} = 29\sqrt{3}$ m. At 30° : $\frac{87}{\tan 30^\circ} = 87\sqrt{3}$ m.

Answer: $87\sqrt{3} - 29\sqrt{3} = 58\sqrt{3}$ m

Q15 A straight highway leads to the foot of a tower. From the top of the tower a car is seen at an angle of depression of 30° , approaching at a uniform speed. Six seconds later the angle of depression is 60° . How much more time does the car take to reach the foot of the tower?

SOLUTION

Let the tower be h m high. The car is first $\frac{h}{\tan 30^\circ} = \sqrt{3}h$ away and then $\frac{h}{\tan 60^\circ} = \frac{h}{\sqrt{3}}$ away.

In 6 seconds it covers $\sqrt{3}h - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$. The remaining distance $\frac{h}{\sqrt{3}}$ is half of this, so at the same speed it takes half the time.

Answer: 3 seconds

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